

Latent Factor Models

CP4285 Modern Recommendation S

CP4285 Instruction Team

18 Aug 2026



Scan for the slides
soc-n.us/cp4285-t2610-w02

01 Announcements and Context

WEEK 01 RECAP

The User–Item Matrix

R	i_1	i_2	i_3	i_4	i_5
u_1	5	?	2	4	?
u_2	?	4	?	1	3
u_3	?	?	?	?	?
u_4	3	?	4	?	2
u_5	?	2	?	5	?
u_6	5	4	5	2	4
u_7	5	1	5	1	5
u_8	?	?	4	?	?

$$R \in \mathbb{R}^{|U| \times |I|}$$

- $u \in U$: user; $i \in I$: item
- r_{ui} : observed feedback
- Ω : observed (u, i) pairs

Warning

A ? is usually **unobserved**, not a negative rating.

To think about

What user types might be represented in the matrix?
What other forms can "ratings" take?

CP4285 - Week 01 - NUS School of Computing

26 / 60

From a Score to a Useful List

Prediction

Estimate a value such as $\hat{r}_{ui}$ for one user–item pair.

Question: "Will this user rate this film highly?"

Top- k ranking

Order candidate items and present a short list.

Question: "Which five films should appear first, now?"

Accurate scores do not automatically make a useful list.

To think about

Does top- k apply to users as well as items?

CP4285 - Week 01 - NUS School of Computing

27 / 60

Similarity I: Pearson Correlation

$$\text{Pearson}(u, v) = \frac{\sum_{k \in I_u \cap I_v} (r_{uk} - \mu_u)(r_{vk} - \mu_v)}{\sqrt{\sum_{k \in I_u \cap I_v} (r_{uk} - \mu_u)^2} \sqrt{\sum_{k \in I_u \cap I_v} (r_{vk} - \mu_v)^2}}$$

Overlap: only mutually rated items contribute.

Centred Rating: rating minus that user's mean, so generous and harsh raters are put on a comparable baseline.

Normalisation: turns aligned variation into a bounded correlation-like score.

Use it when

Rating-scale differences matter.

Watch for

Very small overlap can create an unstable correlation.

CP4285 - Week 01 - NUS School of Computing

42 / 60

From Neighbours to a Prediction

$$\hat{r}_{uj} = \mu_u + \frac{\sum_{v \in P_u(j)} \text{sim}(u, v) (r_{vj} - \mu_v)}{\sum_{v \in P_u(j)} |\text{sim}(u, v)|}$$

Neighbourhood $P_u(j)$: the top- k similar users who rated item j .

Similarity weight: closer neighbours contribute more.

Baseline and deviation: predict a deviation, then add the target user's own mean back.

Pipeline

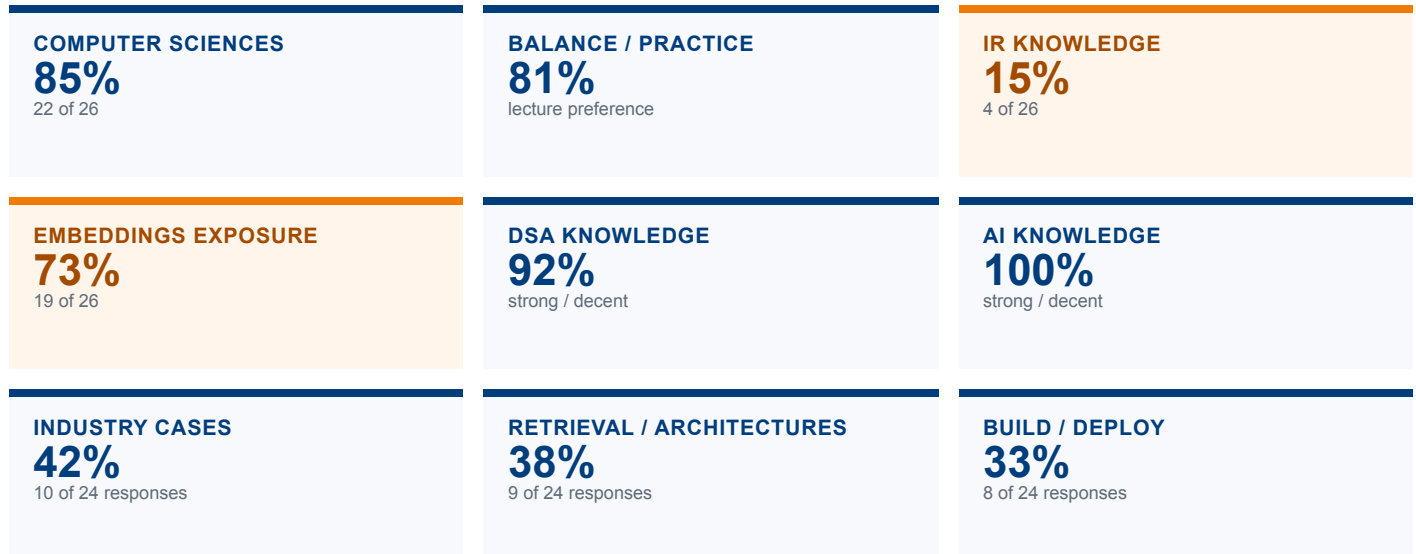
1. choose a similarity
2. select valid top- k neighbours
3. aggregate their evidence
4. rank candidate items by $\hat{r}_{uj}$

CP4285 - Week 01 - NUS School of Computing

55 / 60

Pre-Flight Survey · Metadata Summary

PRE-FLIGHT SURVEY · 26 SUBMISSIONS — cohort signals



Source: Canvas pre-flight survey. Programme selections were multi-select; learning-expectation themes were coded from 24 substantive answers.

Pre-Flight Survey · Voices in the Room

OPEN RESPONSES · REPRESENTATIVE ANONYMOUS PHRASES — larger phrases indicate central themes, not frequency



• Systems and concepts • Building and deployment • Expectations and concerns

Source: Canvas pre-flight survey, anonymous open responses. Phrases are abbreviated from representative excerpts for slide readability.

Essay 1: Micro Case Study: Designing a Production Recommender Surface

Focus

One approved user-facing recommendation surface.



Reason

One or two consequential Week 01–02 choices.



Defend

Evidence, limits, and a substantive suit lens.

-  (Short) **1,500 words** · Due **Mon, 24 Aug 2026, 23:59 SGT**.
- Use no more than **two external sources**; distinguish platform facts from your reasoned proposal.
- Feature a card-suit lens in the technical design, objective, constraint, or evaluation criterion.
- Include an AI-use declaration; AI may assist inquiry but you need to own the argument.

02 From Neighbours to Features to Latent Structure

LAST WEEK PRE-LECTURE EXERCISE**Look for Hidden Structure**

**Canvas discussion board · post before Mon,
23:59 SGT**

Post a short, concrete response to this question; then reply constructively to one classmate.

Name one latent dimension that could explain several ratings—but is not an item metadata label.

Next week: turn this intuition into latent-factor representations of the user-item matrix.

LAST WEEK PRE-LECTURE EXERCISE · CANVAS VOICES

Latent dimensions students noticed

“Quality: Unlike a metadata like price or like count, you can't really quantify quality from the item itself.”



REPLY “Quality is also subjective and can potentially be influenced by trendiness.”

QUALITY

“Nostalgia. It's a property of the relationship between a user and an item which is driven by the user's personal history ...”



REPLY “Could it be modelled by grouping users into distinct age groups?”

“Trust: It is not an item metadata label because it is an underlying, subjective perception that is not directly observable.”



REPLY “The degree of trust cannot be readily quantified and it may vary widely ...”

TRUST

“One latent dimension is how emotionally provocative an item is? Some items naturally stimulate our excitement, motivation or pleasure ...”



REPLY “Psychological emotions are subjective ... and it is tough to quantify.”

EMOTION

Sparse Neighbourhoods Can Fall Silent

Item A	
Item B	
Item C	
Item D	
User 1	
5	
4	
—	
—	
User 2	
—	
5	
4	
—	
User 3	
4	
—	
—	
5	
Target user	
5	
—	
?	
—	

Global Patterns Remain in the Matrix

Neighbourhood CF

- Starts with local overlap.
- Compares named rows or columns.
- Can be silent when overlap is thin.

Latent-factor model

- Learns shared coordinates from many interactions.
- Uses global patterns to score a user–item pair.
- Still inherits the limits of its evidence.



To think about

Does a learned representation make a missing interaction more trustworthy—or only make a prediction possible?

02 Information Retrieval Fundamentals

Text Becomes Searchable as a Vector

Query

feline care

A query is a short text representation of an information need.



Item

description

cat health guide

An item can be represented in the same feature space for comparison.

Information retrieval (IR) asks which available documents or items best satisfy a text-based need.

A Bag of Words Is Sparse and Literal

text / term

cat

feline

health

care

guide

query: feline care

0

1

0

1

0

doc: cat health guide

1

0

1

0

1

- One coordinate per vocabulary term; most coordinates are zero.

- Word order is discarded, but exact lexical evidence is retained.
- **cat** and **fe**l**ine** occupy different coordinates unless we add another mechanism.

TF and IDF Signal Different Things

TF: local evidence

$\text{tf}(t, d)$ = count of term t in document d

Repeated occurrence can strengthen a term's contribution **within one document.**

IDF: collection discrimination

$$\text{idf}(t) = \log \frac{N}{df_t}$$

A term found in almost every document contributes less to distinguishing documents.

$$\text{tfidf}(t, d) = \text{tf}(t, d) \times \text{idf}(t)$$

Worked TF–IDF: Rank a Query Match

Query: machine learning

Term	TF in guide A	IDF	Weight in guide A
machine	2	1.2	2.4
learning	1	1.6	1.6

Query vector

Weights the terms that express the need.



Document vector

Weights the same vocabulary coordinates.



Cosine score

Ranks documents by directional alignment.

A vector-space retriever ranks **lexical evidence**; it has not yet inferred that different words can express a similar idea.

Dense Word Embeddings Learn Neighbourhoods

Training contexts

Words that occur in related contexts provide a learning signal.



Dense vectors

Each word receives a short, continuous learned representation.



Neighbourhoods

Related usage can place vectors closer together.

- Dense vectors have far fewer dimensions than a vocabulary-sized bag-of-words vector.
- They can support semantic generalisation such as related wording.
- Their coordinates are learned patterns, **not human-readable facts or ground truth.**

Sparse and Dense Make Different Trade-offs

Question	TF-IDF	Dense word embedding
What is represented?	Exact vocabulary-term evidence	Learned continuous coordinates
Main strength	Transparent lexical matching	Can generalise across related usage
Main limitation	Synonyms remain separate terms	Learned proximity can be opaque or misleading
Typical score	Cosine similarity over sparse vectors	Cosine/dot-product similarity over dense vectors

Bridge: IR turns text into comparable vectors. LFM will turn interaction history into comparable user and item vectors.



Bandit Time: Where Does TF–IDF Struggle?

- A. TF–IDF forbids any document from receiving more than one score.
- B. *feline* and *cat* are separate vocabulary dimensions unless a mechanism connects them.
- C. Cosine similarity can only compare dense vectors.
- D. IDF removes every rare word from the query.

Answer: . TF–IDF rewards shared lexical coordinates; embeddings can help relate wording, but they introduce a different modelling assumption.

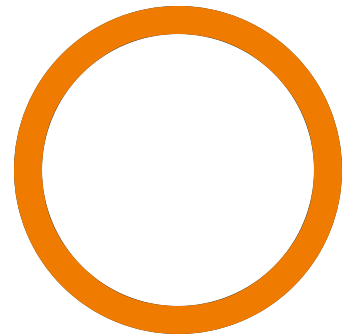
Break — 5 minutes

IR HISTORY · 1958

Keyword search predates the Web

At IBM, **Hans Peter Luhn** demonstrated **Key Word in Context (KWIC)**: an automatic index that rearranged keywords while preserving their textual context.

By the early 1960s, KWIC had become central to hundreds of computerised indexing systems—a useful reminder that retrieval began with finding useful words, not with the Web.



▶ Start
Timer

03 Latent Factor Models

One Vector-Space Idea, Two Evidence Sources

Retrieval	Latent-factor recommendation
Query and document vectors	User and item vectors
Evidence: words in text	Evidence: observed interactions
Score: lexical/semantic match	Score: predicted affinity
Goal: rank relevant documents	Goal: rank plausible items

Analogy, not equivalence: a latent factor is not a word, a recorded metadata field, or a fact about a person.

Factorisation Learns Shared Coordinates

$$R \approx PQ^T$$

Interaction matrix (R)

Observed user–item feedback with many blanks.



User matrix (P)

One learned vector per user.



Item matrix (Q)

One learned vector per item.

A compact set of learned coordinates can capture recurring interaction patterns that local neighbourhoods may miss.

Factors Are Patterns, Not Personal Facts

Tempting shortcut	Better statement
“Dimension 3 is this user’s identity.”	Dimension 3 is a learned coordinate associated with interaction patterns.
“The factor is a hidden genre tag.”	A factor may correlate with metadata, but it is inferred rather than recorded.
“The factor explains the person.”	The factor supports a score under a particular dataset and objective.

A Dot Product Turns Factors into a Score

$$\hat{r}_{ui} = p_u^\top q_i$$

User vector

$$p_u = [p_{u1}, p_{u2}, \dots, p_{uk}]$$

A learned location in the model's preference space.

Item vector

$$q_i = [q_{i1}, q_{i2}, \dots, q_{ik}]$$

How an item aligns with the learned interaction patterns.

The dot product produces a **model score**. A production recommender can still filter, diversify, or re-rank the resulting candidates.

Worked Calculation: Two Items, One Score

$$p_u = [0.8, 0.2], \quad q_A = [0.9, 0.1], \quad q_B = [0.4, 0.8]$$

Item A

Item B

$$\hat{r}_{uA} = 0.8(0.9) + 0.2(0.1) = \mathbf{0.74} \quad \hat{r}_{uB} = 0.8(0.4) + 0.2(0.8) = \mathbf{0.48}$$

Current model verdict: Item A scores higher. **Still unresolved:** diversity, constraints, uncertainty, exposure, and the rest of the candidate list.

Training Fits Evidence, Regularisation Limits It

min

- **Explicit feedback:** a rating can support a prediction-error loss.
- **Implicit feedback:** an interaction supports relative or binary preference evidence.
- **Regularisation:** discourages a model that memorises the observed matrix too conveniently.

BPR Makes LFM Ranking-Aware

$$-\log \sigma(s(u,i) - s(u,j))$$

Observed item (i)

User (u) watched, clicked, or purchased it.



Sampled item (j)

An unobserved alternative for that user.



Pairwise loss

Reward (i) being ranked above (j).

BPR is not a neural architecture. It is a pairwise ranking objective commonly used with LFM-style user and item factors.

Choose Code That Matches the Evidence

Method	Library	Evidence and pedagogical role
Regularised MF / SVD	Surprise SVD	Explicit ratings; prediction-style factor baseline.
BPR-MF	RecBole BPR	Implicit interactions; primary worked example for top- <i>K</i> ranking.
Implicit ALS	<code>implicit</code>	Confidence-weighted sparse implicit feedback; comparison extension.
NMF	scikit-learn or Surprise NMF	Non-negative-factorisation constraint; extension code.
NeuMF	RecBole NeuMF	Neural interaction model; preview

Method	Library	Evidence and pedagogical role
--------	---------	-------------------------------

only, not BPR itself.

Activity 1: Choose the Baseline, Defend the Match

Movie ratings

Users give 1–5 star ratings. Is the score itself the target?

Podcast listens

Users either listened or did not. Which ranked-list outcome matters?

Music discovery

Interactions are implicit and exposure is uneven. What needs monitoring beyond relevance?

Produce: one defensible method/metric pair and one assumption that could make the baseline misleading.

Worked RecBole Example: BPR on MovieLens 100K

```
# test.yaml
USER_ID_FIELD: user_id
ITEM_ID_FIELD: item_id
load_col:
  inter: [user_id, item_id]
embedding_size: 64
epochs: 500
metrics: ['Recall', 'MRR', 'NDCG']
topk: 10

# run.py
from recbole.quick_start import run_recbole
run_recbole(model='BPR', dataset='ml-100k',
            config_file_list=['test.yaml'])
```

What is happening?

- RecBole loads user–item interactions.
- **BPR** learns user/item factors with a pairwise ranking loss.
- **embedding_size** sets the factor dimension.
- Ranking metrics evaluate the declared held-out protocol.

Read Model Output as Limited Evidence

Output artefact	What it can support	What it cannot establish alone
Training loss	Whether the stated loss changes during optimisation.	Real-world user value or fairness.
Recall@10 / NDCG@10	A ranking comparison under the declared held-out protocol.	A production outcome without further evaluation.
Top- <i>k</i> candidates	A prompt for inspection and error analysis.	A causal account of a person's preference.
Learned factors	A representation used by the model.	A verified identity, need, or metadata field.

Scores Need Objectives

Prediction

How close is (r_{ui}) to a numeric target?

Use when a score itself drives a decision; measures can include RMSE or MAE.

Ranking

Do useful items appear near the top of the visible list?

Use top- k measures such as NDCG@10 or Recall@10 for the declared task.

A latent-factor score is not enough: the **objective and metric decide what “better” means.**



Bandit Time: What Should a Home Screen Optimise?

- A. RMSE over predicted ratings.
- B. NDCG@10.
- C. The number of metadata tags per item.
- D. The age of the user account.

Answer: . NDCG@10 is position-sensitive; it assesses useful ordering in a displayed top- k .

History Is Evidence, Not Preference

Historical process	Model consequence	Question to ask
Uneven exposure	Sparse or distorted interaction evidence	Who had a genuine chance to interact?
Popularity reinforcement	Popular items receive stronger learned signals	Who receives repeated visibility?
Re-ranking for discovery	Today's policy changes tomorrow's data	Who gets a credible opportunity to be seen?

Use historical behaviour as evidence with limits—not as a neutral record of preference.

E1 Bridge: Choose the Consequential Decision

For one approved recommendation surface, make a focused argument around one connected choice:

Choice	Example question
Representation	Why use lexical features, latent factors, or both?
Objective	Why optimise a ranked list rather than numeric prediction?
Method	Why is BPR-MF, explicit MF, ALS, or another baseline appropriate?
Risk	How could missing exposure distort the interaction evidence?

NEXT WEEK PRE-LECTURE EXERCISE

When two lists rank similarly, what should decide?

Suppose two ranked lists have similar NDCG@10, but one concentrates attention on already popular items while the other broadens credible discovery.

Post one diagnostic you would use, explain what it reveals, and reply constructively to one classmate.

Summary



Representations Enable; Objectives Decide

- **TF-IDF** represents lexical evidence; **dense embeddings** can generalise across learned contextual patterns.
- **LFM** represents user–item interaction patterns as learned factors—not identities or recorded metadata.
- **BPR** keeps the latent-factor idea but trains it for relative ranking under implicit feedback.
- A score or offline metric is useful evidence, never a complete account of user value or fairness.



Better representation helps; responsible objectives and evaluation determine what the system actually delivers.



CP4285 · Week 02 · NUS School of Computing

Speaker notes